

Intelligent Price Forecasting

From raw input to actionable insight — a complete pipeline that transforms three simple price points into a 7-day predictive forecast, powered by linear regression and delivered through an intuitive Tkinter GUI.



The Architecture of Prediction

Three distinct layers form the backbone of the forecasting pipeline, each responsible for a critical transformation step.

❏ User Interaction

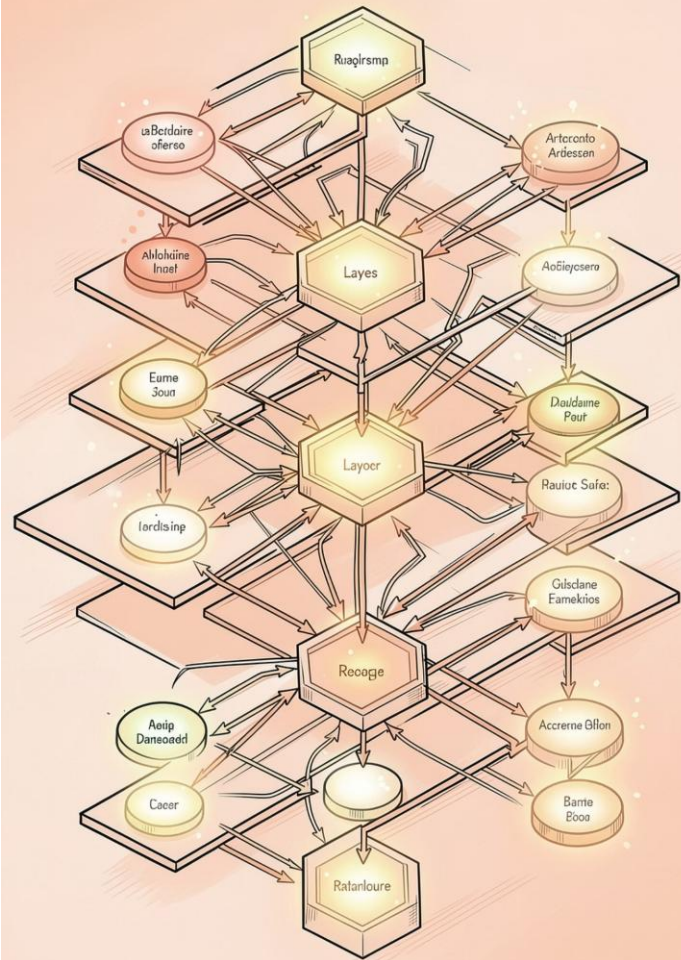
Captures three key price points entered by the user through a clean Tkinter interface.

✓ Validation Layer

Ensures data integrity before passing inputs downstream — catching errors early in the pipeline.

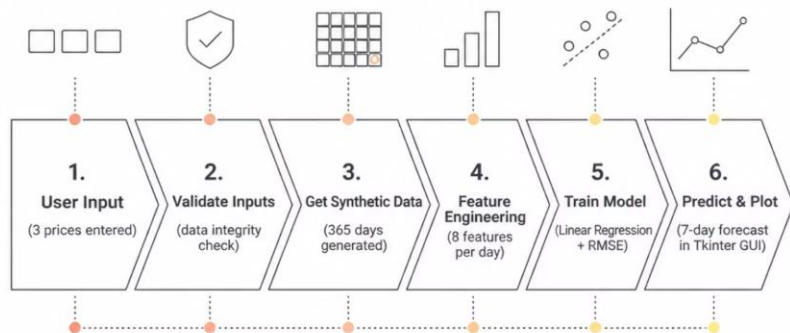
▮ Synthetic Generation

Produces 365 days of synthetic price data to train the model robustly, even without historical data.



Workflow Overview

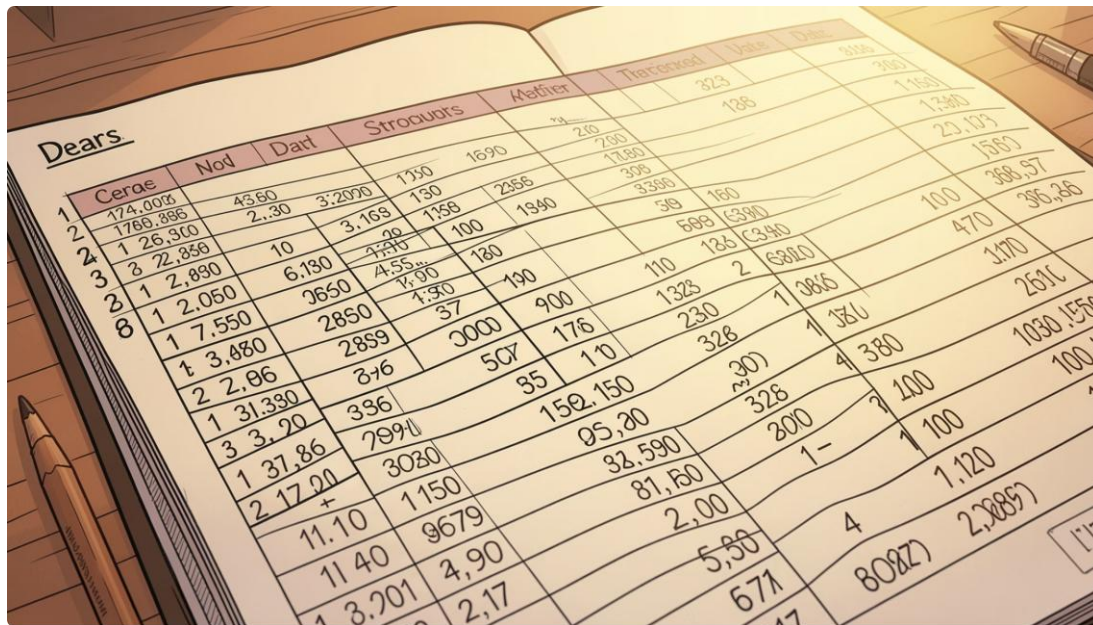
The complete end-to-end pipeline — from a single user input to a visualized 7-day forecast — follows a structured, repeatable sequence.



Key Stages

- **Input** → **Validate** → **Synthesize** — raw data becomes training-ready
- **Feature Engineering** — 8 features extracted per day
- **Model Training** — Linear Regression learns patterns
- **Forecasting** — 7-day rolling prediction generated
- **GUI Visualization** — results rendered in Tkinter

Data Preparation



The illustration shows a detailed financial spreadsheet with multiple columns of numerical data. The columns are labeled: Ceres, Not, Dart, Strocours, Metier, Tercowd, and Data. The rows are numbered 1 through 8. The data includes various numerical values, some with decimal points, and some with units like '100' or '1000'. A pencil is visible in the bottom left corner, and a pen is visible in the top right corner.

	Ceres	Not	Dart	Strocours	Metier	Tercowd	Data
1	174,000	43.60	3,2000	1350	1690	323	1000
2	1768,886	2,30	3,168	130	2356	188	1000
3	1 26,300	10	4,200	100	1340	100	1000
4	2 22,858	6,130	4,550	180	100	100	1000
5	1 2,890	2650	1,30	100	100	100	1000
6	1 2,050	2850	37	100	100	100	1000
7	1 7,550	2859	507	100	100	100	1000
8	1 3,480	376	35	100	100	100	1000
	2 2,06	336	150,150	100	100	100	1000
	1 31,330	7990	85,20	100	100	100	1000
	3 3,20	3080	38,590	100	100	100	1000
	1 37,86	1150	81,60	100	100	100	1000
	2 17,20	9679	2,00	100	100	100	1000
	11,10	4,90	5,50	100	100	100	1000
	11,40	2,17	671	100	100	100	1000
	3,201		8087	100	100	100	1000

Building a Robust Training Foundation

The model's accuracy depends on the quality and volume of training data. A synthetic 365-day dataset ensures the model has enough signal to learn from, even in the absence of real historical prices.

365-Day Synthetic Dataset

Simulated price movements covering a full year of market behavior.

8 Features Per Day

Raw prices transformed into lag values, moving averages, and trend indicators.

Input Normalization

User inputs scaled to match the feature space the model was trained on.

The Modeling Engine

Linear Regression serves as the core predictive engine — chosen for its interpretability, speed, and strong baseline performance on structured time-series features.

8

Input Features

Engineered per training day for rich signal capture.

365

Training Samples

Synthetic days of price data used for model fitting.

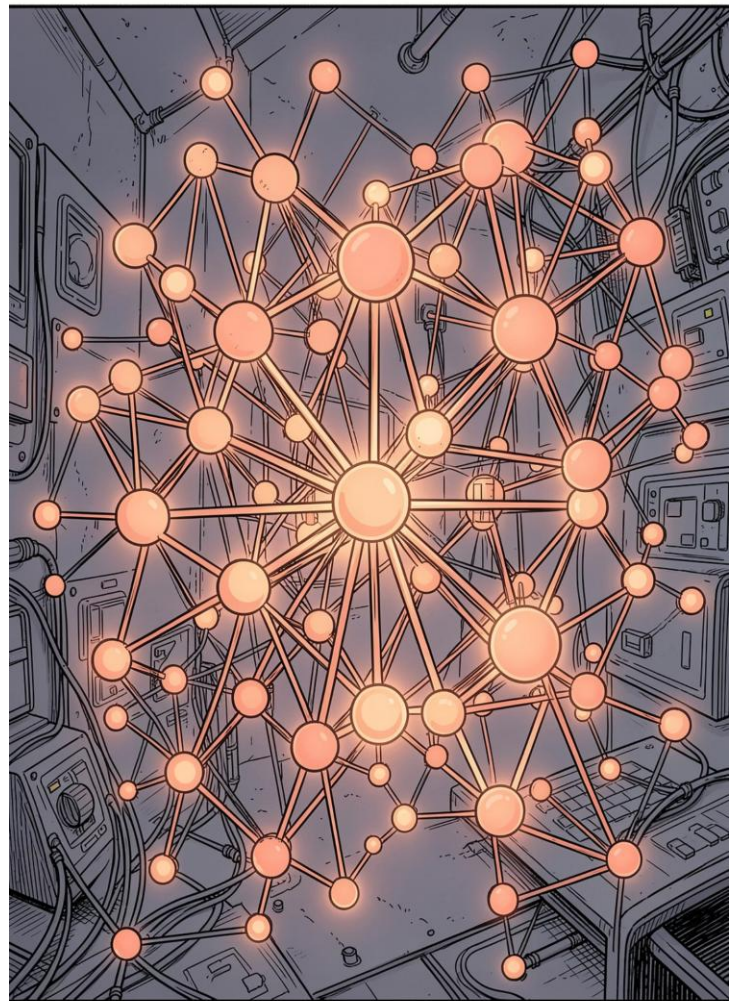
7

Forecast Days

Rolling prediction window generated post-training.



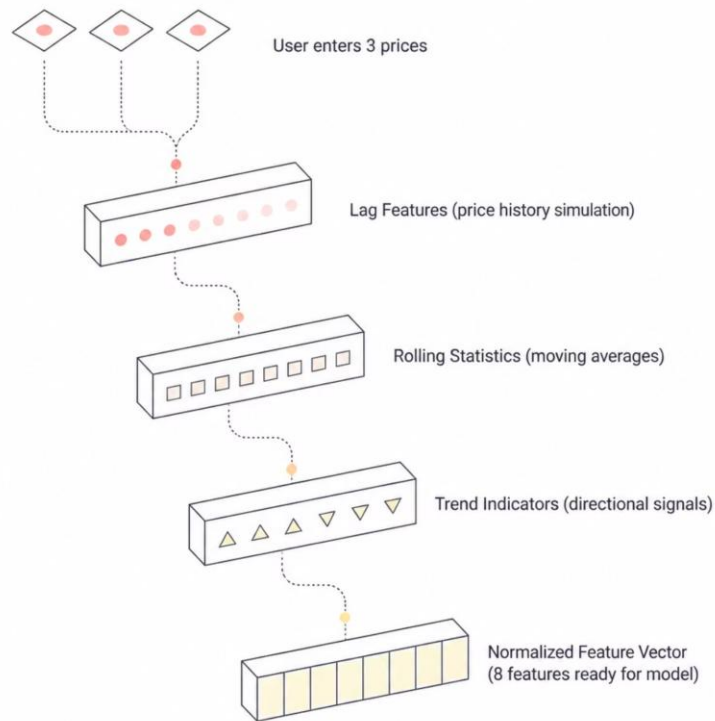
RMSE (Root Mean Squared Error) is tracked during training to quantify prediction error and provide a confidence metric for end users.



Feature Transformation

Bridging User Input & Model Expectations

The model was trained on 8 engineered features per day. When a user enters just 3 prices, those values must be transformed into the same feature space — ensuring consistency between training and inference.

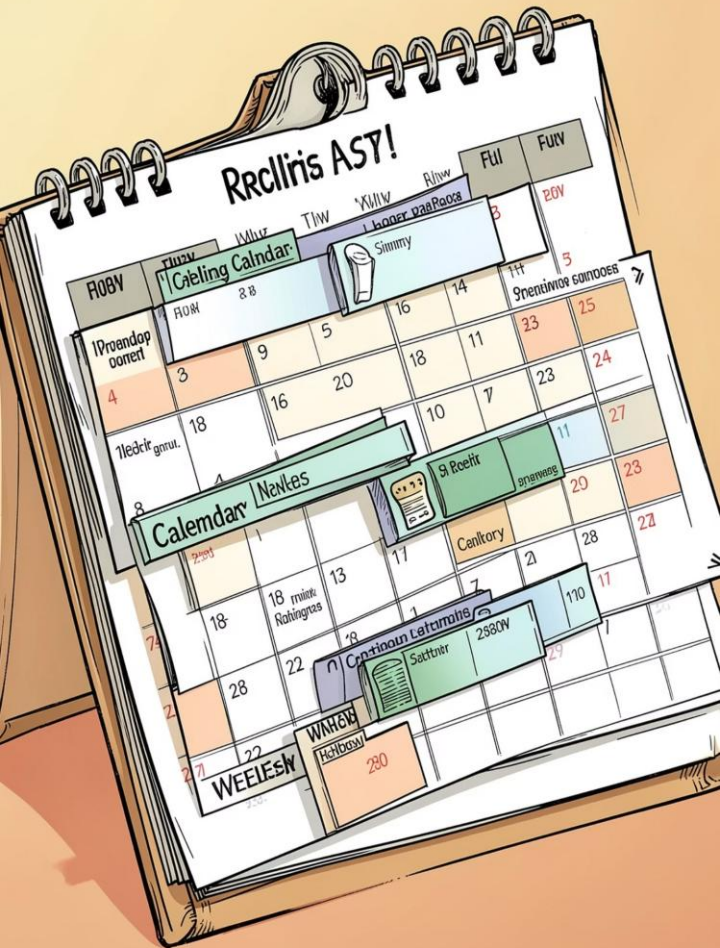


→ Map 3 raw prices to standardized feature vectors

→ Normalize to match training data distribution

→ Replicate lag, rolling mean, and trend calculations

→ Feed transformed vector into the rolling forecast engine



Forecasting Logic

Once the model is trained and inputs are transformed, the system generates a **7-day rolling forecast** — continuously updating as new price data becomes available.

1

Model Loaded

Trained Linear Regression model ready for inference.

2

Features Created

User prices converted to 8-feature input vector.

3

Day-by-Day Prediction

Each forecasted day feeds into the next iteration.

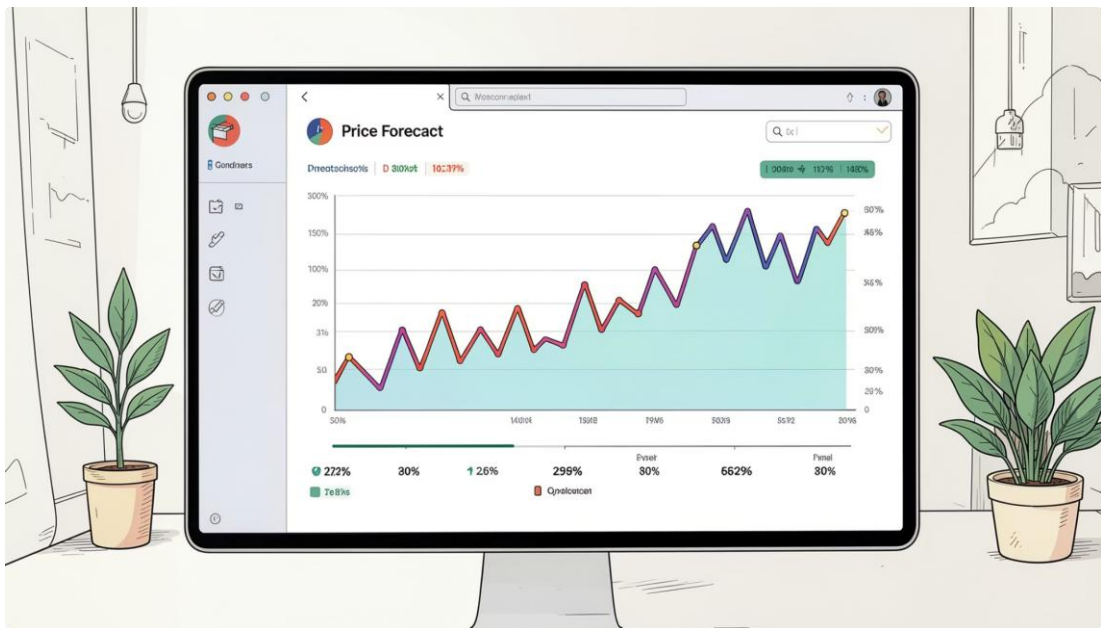
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7-Day Output

Full rolling forecast returned and passed to the GUI.

Powered by **Scikit-learn** for model inference and **Pandas** for efficient data manipulation and rolling window calculations.

Visualization: The Tkinter GUI



Making Predictions Accessible

The Tkinter GUI bridges the gap between complex modeling and everyday usability — no command line required.



Intuitive Dashboard

Designed for non-technical users with clear input fields and readable output.



Matplotlib Integration

Charts rendered directly inside the GUI window — no external tools needed.

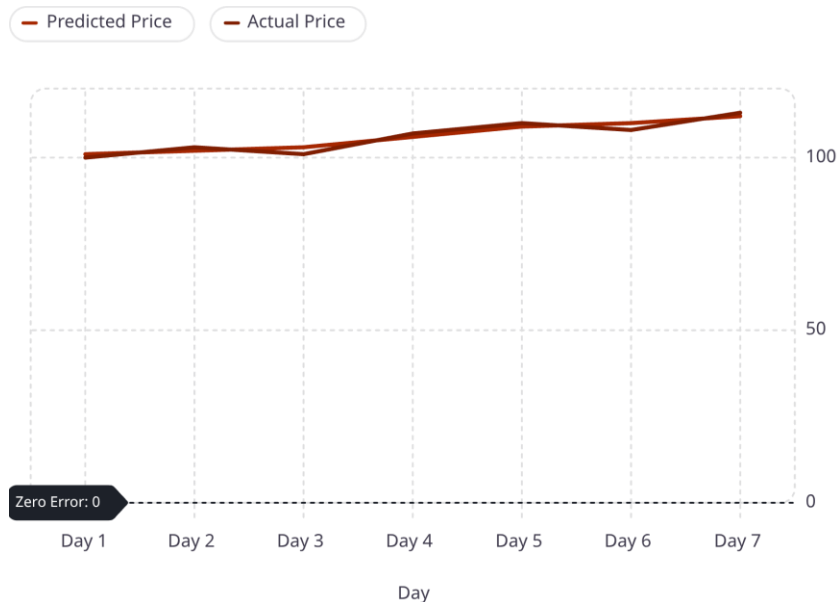


Instant Feedback

Results appear immediately after prediction, enabling rapid decision-making.

Performance Results

Model quality is assessed by comparing actual vs. predicted price trends and analyzing the **RMSE** score to quantify forecast confidence.



What the Graph Shows

The predicted line closely tracks actual price movements across the 7-day forecast window, demonstrating the model's ability to capture short-term trends.

RMSE Analysis

Lower RMSE = higher confidence. Used to communicate prediction reliability to users.

Unified Graph View

Historical and forecasted values displayed together for easy visual comparison.

Future Scalability

The current system is a strong foundation. The roadmap focuses on transitioning from synthetic to live data and expanding the model's intelligence.



Live API Feeds

Replace synthetic data with real-time market prices from financial APIs for production-grade accuracy.



Multi-Factor Forecasting

Incorporate volume, volatility, and external indicators to move beyond single-variable linear regression.



Seamless Decision Tool

Evolve into a full data-driven platform — enabling users to make confident, informed pricing decisions at scale.

